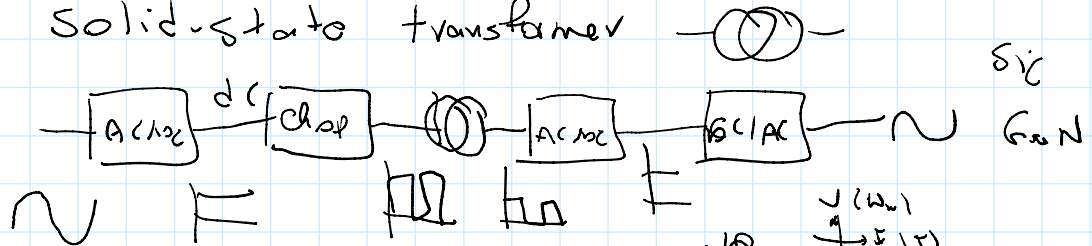


DC-DC converters

→ DC choppers (DC motor control)
 → switching power supplies (chopper + filter)
 high frequency → magnetic elements size is reduced.

Solid-state transformer

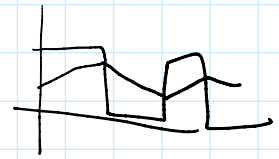


DC choppers → step-down

1Q dc chopper

$$V_o = D_s V_i$$

$$P_s = \frac{V_o I_o}{T_s}$$



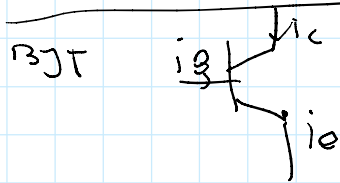
enout

approximate 1

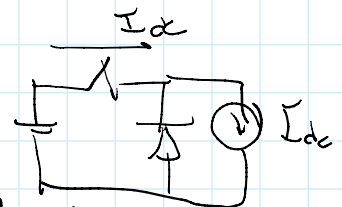
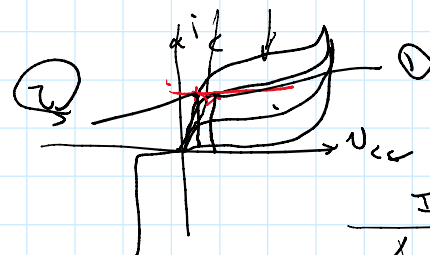
$$\bar{I}_o \geq \frac{\Delta I_o}{2}$$

Semiconductor switches used in DC/DC converters

fully controlled
 current controlled
 voltage =

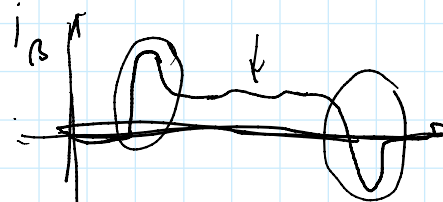


$$i_c = \beta i_b$$



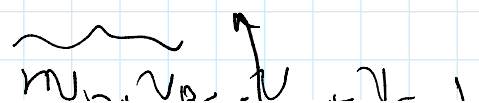
$$i_b = \frac{i_c}{\beta} \quad i_b > i_c / \beta \Rightarrow \text{hard saturation}$$

ideal i_b for a BJT

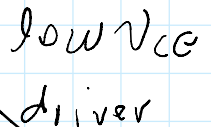
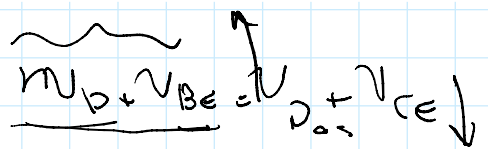


$$\text{hard saturation} \Rightarrow i_b > \frac{i_c}{\beta}$$

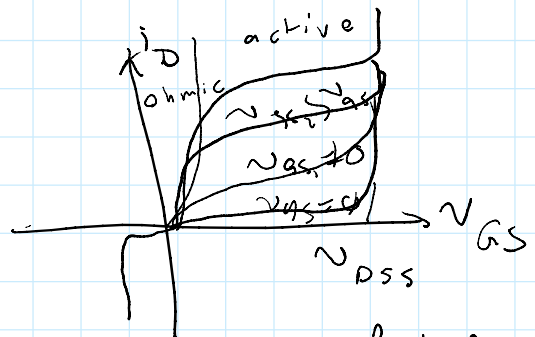
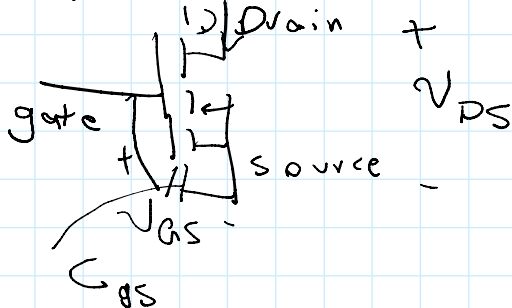
B-Knorr Diagram



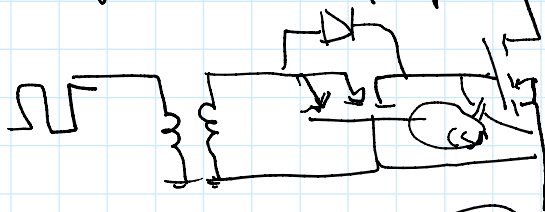
- Baker's clamp



power mosfet:



Driver for a power MOSFET

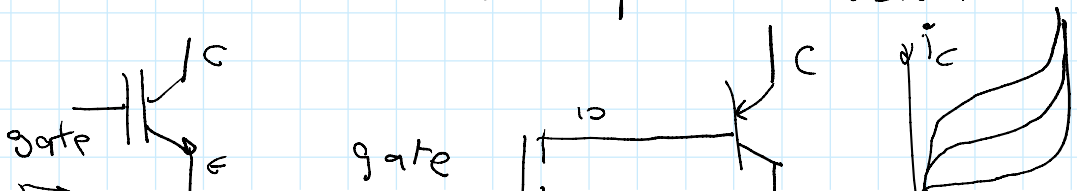


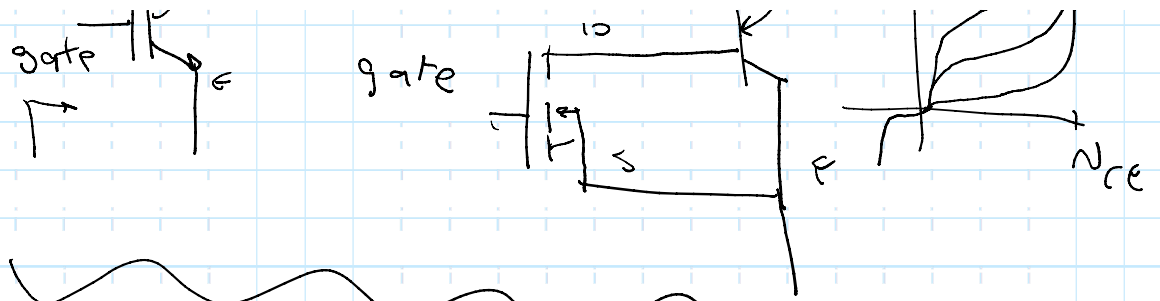
- high frequency switching

- high V_{DS}

IG BT:

Insulated Gate Bipolar Transistor

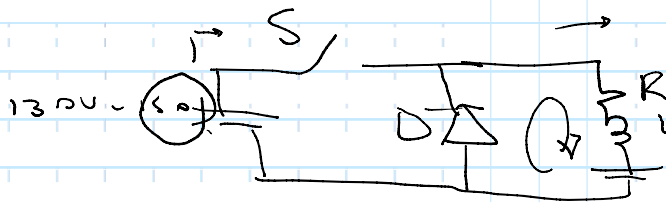




PC-PC Example:

A dc motor 10kW - 120V
 $R_a = 0.5 \Omega$, $L_a = 40 \mu H$
 $f_s = 5 \text{ Khz}$, $P_{\text{min}} = 500W$
 $V_{in} = 130V - 150V$

a) draw the circuit



IGBT $\left\{ \begin{array}{l} V_{CS(15V)} \checkmark \\ I_{C(10A)} \checkmark \\ f_s = 5 \text{ Khz} \end{array} \right.$
 $V_{CE(sus)} \geq 150 \times 5 \text{ s.f.} = 225V$

$\left\{ \begin{array}{l} P_{\text{out}} = 10 \text{ kW} \\ 120V \end{array} \right.$

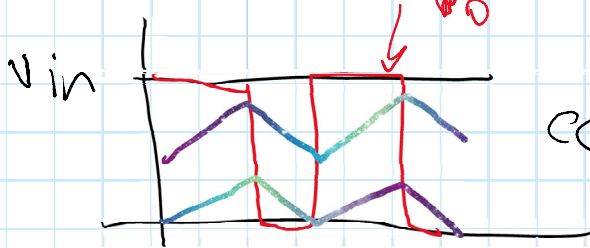
lossless system
 $P_{in} = P_{out}$

$$I_C \geq \frac{10,000}{130} \approx 77A$$

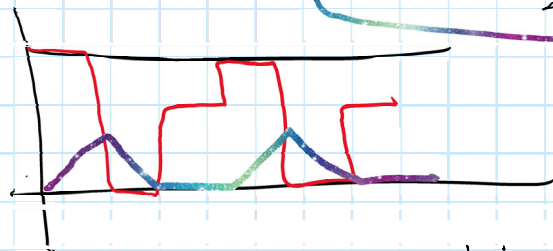
b) does the converter operate in a ccm?
 always

$$\bar{I}_{C \text{ min}} \geq \frac{\Delta I}{2}$$

$$\bar{I}_O \geq \frac{\Delta I_O}{2}$$



CCM

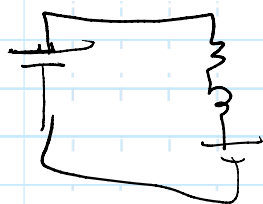


DCM

$$\bar{I}_O \geq \frac{\Delta I_O}{2}$$

(11)

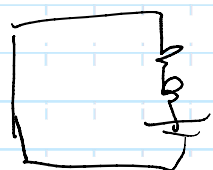
s on



$$Ri_O + L \frac{di_O}{dt} + E = V_i$$

$$0 \leq t < t_{on}$$

s off



$$Ri_O + L \frac{di_O}{dt} + E = 0$$

$$t_{off} \leq t < t_{off} + t_{on}$$

$$di_O = 0$$

$$s \text{ on} \Rightarrow \left| \frac{di_O}{dt} \right|_{\text{on}} = \frac{V_i - Ri_O - E}{L}$$

$$\leq \text{or} \Rightarrow \left| \frac{D(1-D)}{2f_s} \right| \left| \frac{V_i}{L} \right| = \frac{V_i - V_o - E}{L}$$

$$\leq \text{or} \Rightarrow \left| \frac{di_o}{dt} \right| = \frac{D(1-D)}{2f_s} = \frac{R \tilde{I}_o + E}{L}$$

$$|D(1-D)| = |D(1-D)| \Rightarrow \tilde{I}_o = \frac{D(1-D)V_i - E}{R}$$

$$|D(1-D)| = \frac{D(1-D)V_i}{2Lf_s} \quad \tilde{I}_o = \tilde{I}_{in} = \frac{P_o}{V_o} \quad (\pi)$$

$$\tilde{I}_o \geq \frac{D(1-D)V_i}{2Lf_s} \Rightarrow \frac{P_o}{V_o} \geq \frac{D(1-D)V_i}{2Lf_s} \quad \frac{V_o}{V_i} = D$$

$$\frac{P_o}{V_o} \geq \frac{D(1-D)V_o}{2Lf_s} \Rightarrow Lf_s \geq \frac{(1-D)V_o^2}{2P_{o,min}} \quad \text{CRM}$$

$$V_i = 130V - 150V$$

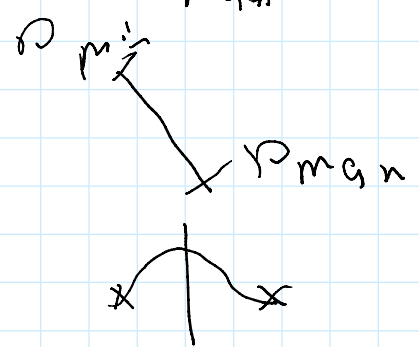
$$V_o = 120V \Rightarrow 0.8 \leq D \leq 0.923$$

$$\frac{D(1-D)}{2D}$$

$$D \leq 0.8 \Rightarrow L \geq 0.6 \text{ mH}$$

$$D \leq 0.923 \Rightarrow L \geq 0.2 \text{ mH} \Rightarrow \text{PCM}$$

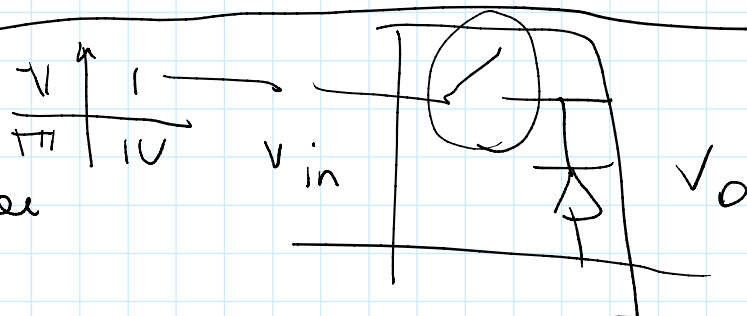
$$Lf_s < \frac{(1-D)V_o^2}{2P_{o,max}^2} \quad \text{PCM}$$



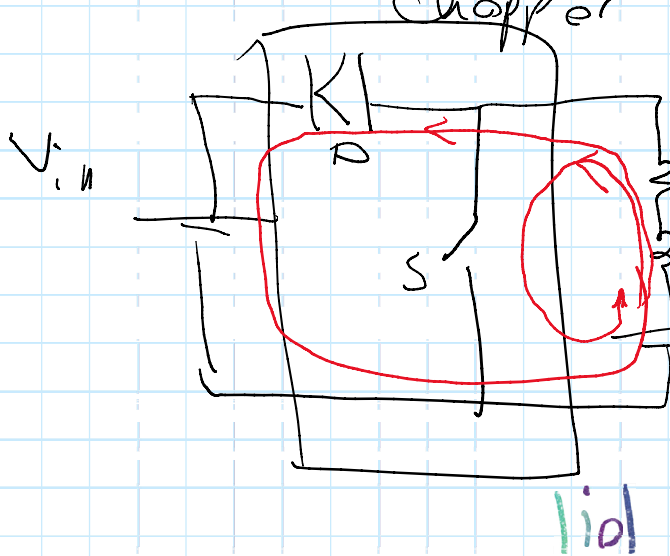
1st Chopper

1st - quadrant chopper

2nd - quadrant



2nd quadrant v
chopper



$\begin{cases} S \text{ on} \\ D \text{ off} \end{cases} \rightarrow V_o = 0, |i_o| \uparrow$

$\begin{cases} S \text{ off} \\ D \text{ on} \end{cases} \rightarrow V_o = V_i, |i_o| \downarrow$



$P < 0$ — brake

$$V_o \approx \frac{t_{on} \times 0 + t_{off} \times V_i}{T_s}$$

$$V_o = \frac{t_{off}}{T_s} V_i = (1 - D_s) V_i$$

$$M = \frac{V_o}{V_i} = 1 - D_s \quad \oplus$$

$$m = \frac{V_o}{V_i} = D_s \quad \ominus$$